

BEEP CODE ERROR

BEEP CODE ERROR

- One Short beep during power on
 - No error during POST
- Long beeps in endless loop
 - No RAM installed or detected
- One long beep followed by three short beeps
 - No VGA card detected or bad video RAM
- High frequency beeps during operation
 - CPU overheated

BIOS

- stands for the Basic Input/Output System.
- (pronounced "bye-ose")
- BIOS in part refers to the firmware code run by a PC when first powered on
- The primary function of the BIOS is to identify and initialize system component hardware (such as the video display card, hard disk, and floppy disk) and some other hardware devices.
- Prior to the early 1990s, BIOSes were stored in ROM or PROM (**programmable read-only memory**) chips, which could not be altered by users.
- As its complexity and need for updates grew, and re-programmable parts became more available, BIOS firmware was most commonly stored on EEPROM (**Electrically Erasable Programmable Read-Only Memory**) or flash memory devices



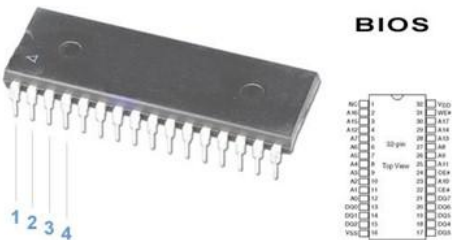
Types of BIOS Packaging

- DIP - **dual in-line package (DIP)**, sometimes called a **DIL** package
- is an electronic device package with a rectangular housing and two parallel rows of electrical connecting pins, usually protruding from the larger sides of the package and bent downward.
- PLCC - **Plastic Leaded Chip Carrier**
- is a four-sided "J"-leaded plastic integrated circuit package with pin spacings of 0.05" (1.27 mm)

DIP (Dual In-Line Package) Chip



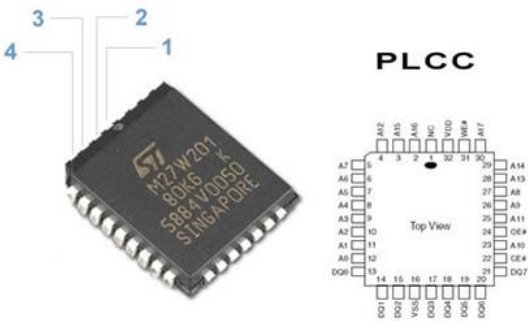
DIP 32



BIOS



PLCC Chip (Plastic Leaded Chip Carrier)



PLCC



Major BIOS Manufacturer

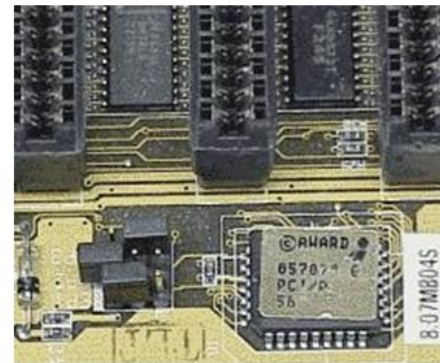
AMI BIOS Beep Codes

- 1 short - power-on self test successful
- 1 long, 8 short - Video adapter error
- 3 short - Memory error



AWARD BIOS Beep Codes

- 1 long, 3 short – Video adapter error
- Repeating (endless loop)
 - Memory error



Compaq BIOS Beep Codes

- 1 long, 2 short - Video error
- 1 short, 2 long - Bad RAM



BIOS

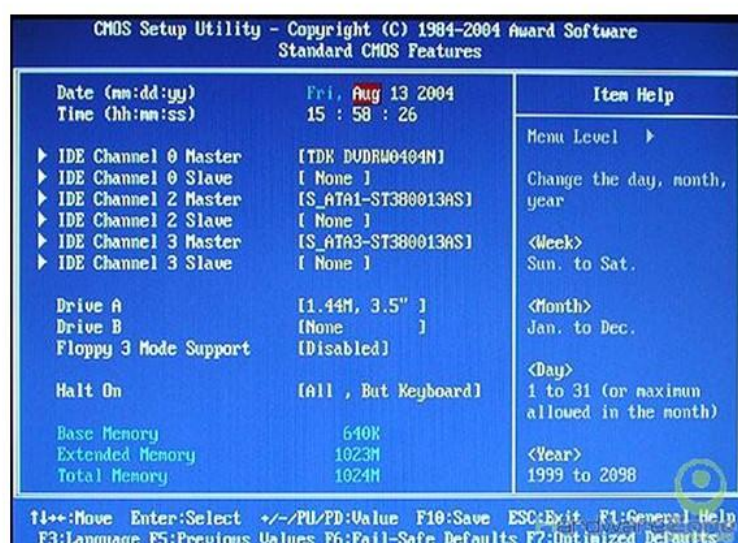
(Basic Input Output Devices)

CONFIGURATION



Standard CMOS Features

- Here you can setup the basic BIOS features such as date, time, type of floppy etc. You can specify what IDE devices you have such as Hard drive, CD-ROM drive etc.
- The easiest way to setup the IDE devices is by leaving it set to auto. This allows the BIOS to detect the devices automatically so you don't have to do it manually.
- At the bottom, it also displays the total memory in your system.



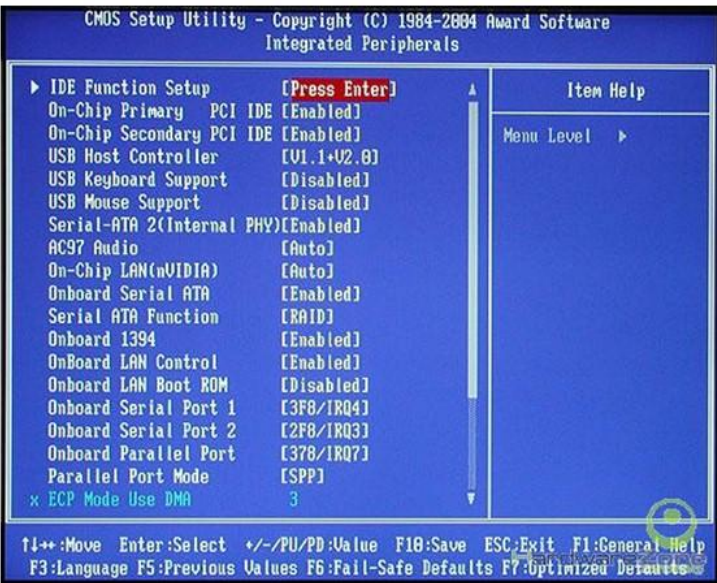
Advanced BIOS Features

- This menu lets you change system settings such as boot sequence, character typing input rates, CPU cache settings, security options, and virus warning settings.
- Lets you select the primary boot device, or the order in which devices will be booted. The default is generally A:, then C:, but newer motherboards let you boot from other hard drives



Integrated Peripherals

- This section of the BIOS setup program contains settings for built-in peripherals included with the motherboard. This includes serial and parallel ports, as well as audio, LAN, and USB ports.
- These settings are used to enable or disable integrated peripheral support, and set the various resources they use.



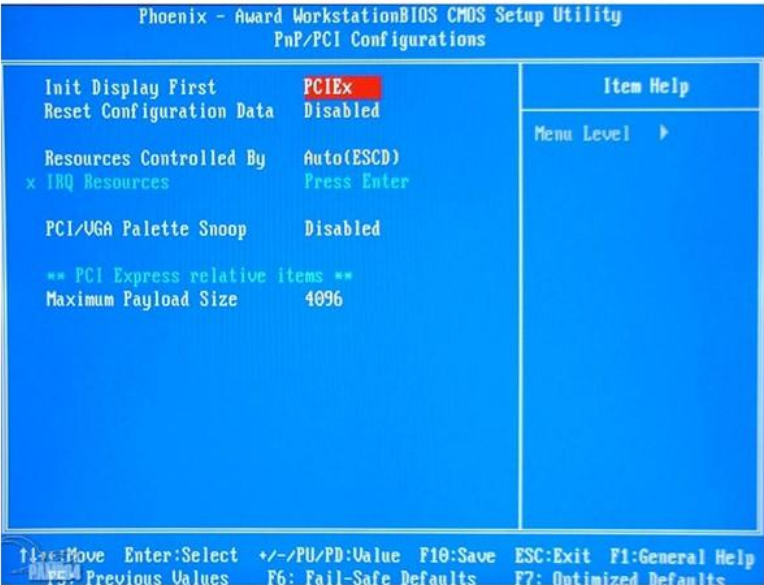
Power Management Setup

- The Power menu items allow you to change the settings for the Advance Configuration and Power Interface (ACPI)
- You'll be able to set up how you want your computer to be suspended and resumed.



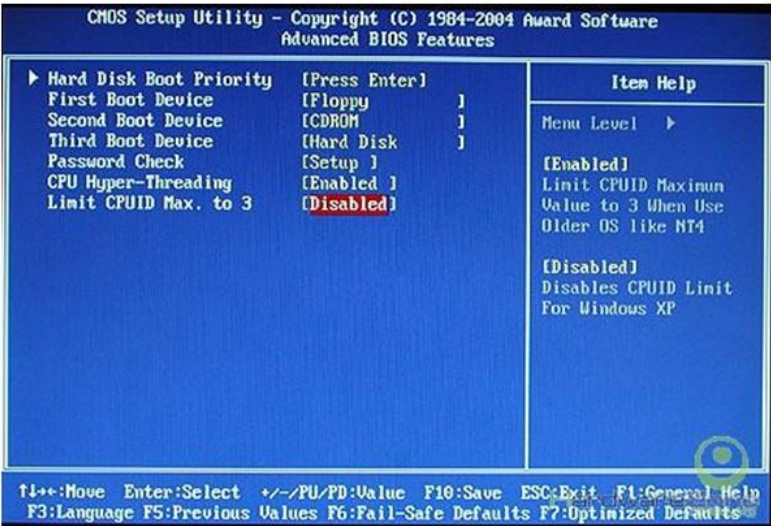
PnP / PCI Configuration

- The PnP/PCI Configuration section of the BIOS controls the settings for the motherboard's PCI slots. This includes the plug-and-play capability of the BIOS.
- If you are installing a Plug-n-Play operating system such as *Microsoft windows* then the PNP OS Installed should read YES.
- If you are installing another operating system such as *Linux* then it should read NO.
- You can leave the rest like they are.



Advanced BIOS Features

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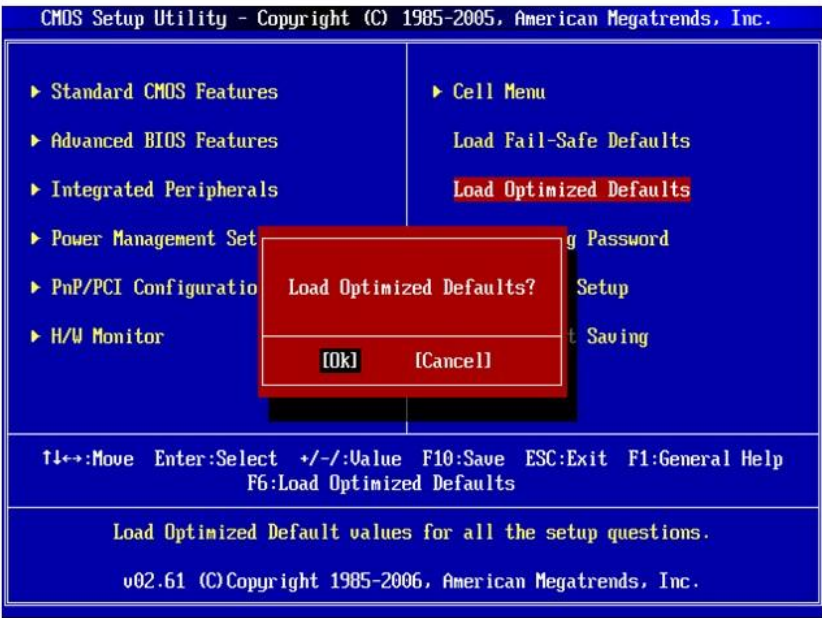
LOAD FAIL-SAFE DEFAULTS

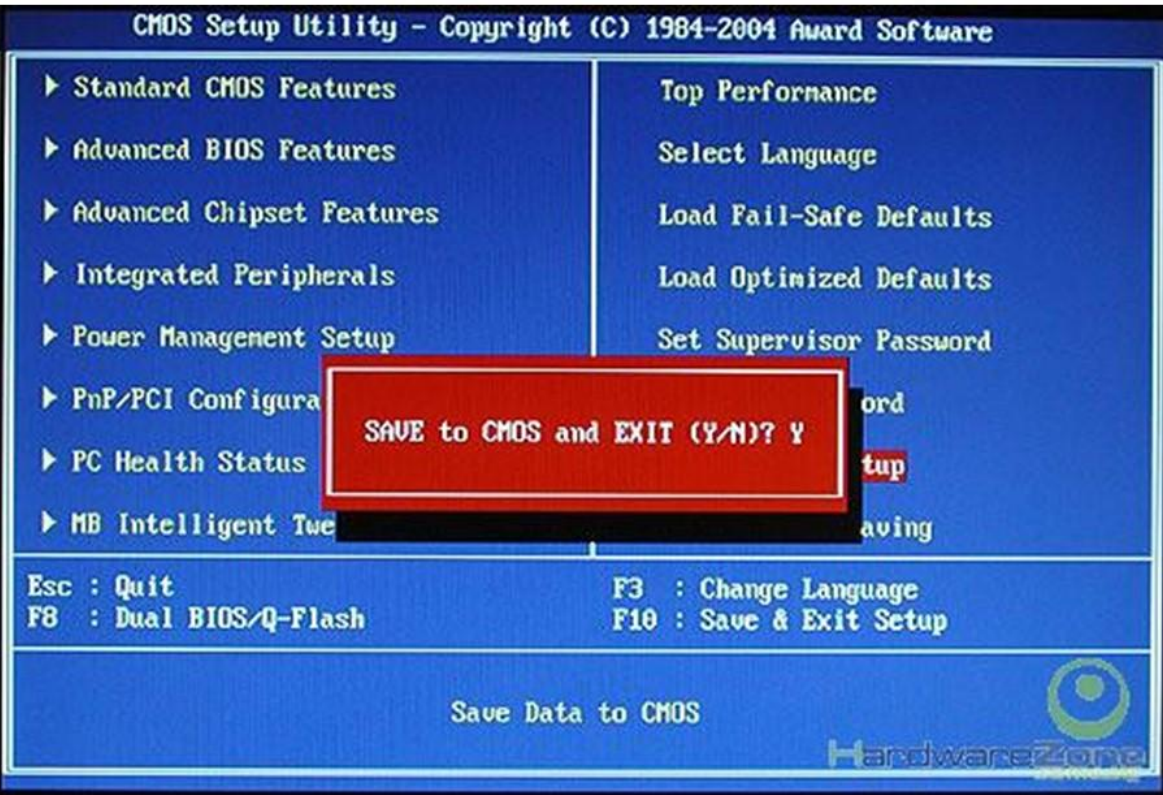
- If you made changes to the BIOS and your system becomes unstable as a result, you can change it back to default.
- However if you made many changes and don't know which one is causing the problem, your best bet is to choose the option "Load Fail Safe Mode Defaults" from the BIOS menu.
- This uses a minimal performance setting, but the system would run in a stable way.
- From the dialog box Choose "Y" followed by enter to load Fail-Safe Defaults.



Load Optimized Defaults

- Like the Fail-Safe mode above, this option loads the BIOS default settings, but runs the system at optimal performance.
- From the dialog box Choose "Y" followed by enter to load Optimized Defaults.





Partitioning

Partitioning

- is the creation of divisions of a hard disk
- Once a disk is divided into several partitions, directories and files can be grouped by categories such as data type and type usage
- A PC hard disk can contain either as many as **four** primary partitions, or 1-3 primaries and a single extended partition



FILE SYSTEM

FILE SYSTEM

- is a method for storing and organizing computer files and the data they contain to make it easy to find and access them
- File systems may use a data storage device such as a hard disk or CD-ROM

FAT

- **File Allocation Table**
- is a computer file system originally developed by [Bill Gates](#) and [Marc McDonald](#) in 1976/1977
- There are versions called FAT12, FAT16, and FAT32
- The purpose of this table is to keep track of which areas of the disk are available and which areas are in use

FAT12

- initial version of FAT
- also called 12-bit FAT
- designed as a file system for floppy diskettes
- most suitable for very small volumes, and is used on floppy disks and hard disk partitions smaller than about 16 MB

FAT16

- File allocation table that uses 16 bits
- Commonly used with DOS and Windows 95 systems
- The ultimate capacity of a FAT16 partition is 2 GB

FAT32

- originally introduced in Windows 95 Service Pack 2
- supports drives up to 2 terabytes in size
- cannot format a volume larger than 32 gigabytes (GB) in size during the Windows XP installation process
- does not allow compression using DriveSpace
- None of the FAT file systems provide the file security, compression, fault tolerance, or crash recovery abilities that NTFS does
- FAT32 volumes are not accessible from any other operating systems other than Windows 95 OSR2 and Windows 98

NTFS

- New Technology File System
- use of advanced data structures to improve performance, reliability, and disk space utilization
- It provides security by supporting access control and ownership privileges, meaning you can set permission for groups or individual users to access certain files
- is a recoverable file system, meaning it has the ability to undo or redo operations that failed due to such problems as system failure or power loss
- also provides Disk quotas, Encryption, supports Macintosh files and compression of individual files and folders

The Advantages of NTFS

- It provides for greatly increased security
- file-by-file compression
- Quotas
- encryption

Release date	Product name	Version
November 1985	Windows 1.01	1.01
November 1987	Windows 2.03	2.03
March 1989	Windows 2.11	2.11
May 1990	Windows 3.0	3.0
March 1992	Windows 3.1	3.1
October 1992	Windows For Workgroups 3.1	3.1
July 1993	Windows NT 3.1	NT 3.1
December 1993	Windows For Workgroups 3.11	3.11
January 1994	Windows 3.2 (released in Simplified Chinese only)	3.2
September 1994	Windows NT 3.5	NT 3.5
May 1995	Windows NT 3.51	NT 3.51
August 1995	Windows 95	4.0.950
July 1996	Windows NT 4.0	NT 4.0
June 1998	Windows 98	4.10.1998
May 1999	Windows 98 SE	4.10.2222

February 2000	Windows 2000	NT 5.0.3700.6690
September 2000	Windows Me	4.90.3000
October 2001	Windows XP	NT 5.1.2600
March 2003	Windows XP 64-bit Edition 2003	NT 5.2.3790
April 2003	Windows Server 2003	NT 5.2.3790
April 2005	Windows XP Professional x64 Edition	NT 5.2.3790
July 2006	Windows Fundamentals for Legacy PCs	NT 5.1.2600
November 2006 (volume licensing)/January 2007 (retail)	Windows Vista	NT 6.0.6000
July 2007	Windows Home Server	NT 5.2.4500
February 2008	Windows Server 2008	NT 6.0.6001
2010 (planned)	Windows 7 (codenamed Blackcomb, then Vienna)	NT 7.0

Operating System Requirements

Windows 95

- Personal computer with a 386DX or higher processor (486 recommended)
- 4 megabytes (MB) of memory (8 MB recommended)
- Typical hard disk space required to install Windows 95 on a clean system: 50-55 MB
The actual requirement varies depending on the features you choose to install.
- One 3.5-inch floppy disk drive



Windows 98

- A personal computer with a 486DX 66 megahertz (MHz) or faster processor (Pentium central processing unit recommended)
- 16 megabytes (MB) of memory (24 MB recommended).
- A full install of Windows 98 requires 175 MB of free hard disk space, but may range from between 140 MB and 255 MB
- One 3.5-inch high-density floppy disk drive.



Windows 2000

Windows 2000 Professional:

- 133 MHz or higher Pentium-compatible CPU
- 32 megabytes (MB) of RAM minimum (64MB recommended)
- 700 megabytes hard disk space (2GB recommended)

Windows 2000 Server:

- 133 MHz CPU
- 256 MB of RAM minimum
- 2 GB hard disk space

Windows 2000 Advanced Server:

- 133 MHz CPU
- 256 MB of RAM recommended minimum
- 2 GB hard disk space



Windows ME

System requirements

Minimum system requirements:

Processor: Pentium 150MHz

Hard drive space: 320MB

RAM: 32MB

Recommended system requirements:

Processor: Pentium II 300MHz

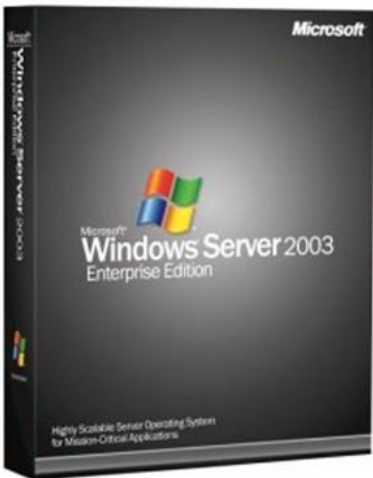
Hard drive space: 2GB

RAM: 96MB



Windows 2003 Server

- 133-MHz or faster processor
- 128 MB of RAM minimum required
- 1.5 GB of available hard-disk space
- CD-ROM or DVD-ROM drive



Windows Xp



	Minimum	Recommended
Processor	233 MHz	300 MHz or higher
Memory	64 MB RAM (may limit performance and some features) ^[20]	128 MB RAM or higher
Video adapter and monitor	Super VGA (800 x 600)	Super VGA (800 x 600) or higher resolution
Hard drive disk free space	1.5 GB	1.5 GB or higher
Drives	CD-ROM drive or DVD drive	CD-ROM drive or DVD drive
Devices	Keyboard, Microsoft Mouse or compatible pointing device	Keyboard, Microsoft Mouse or compatible pointing device
Others	Sound card, Speakers or headphones	Sound card, Speakers or headphones

Windows Vista

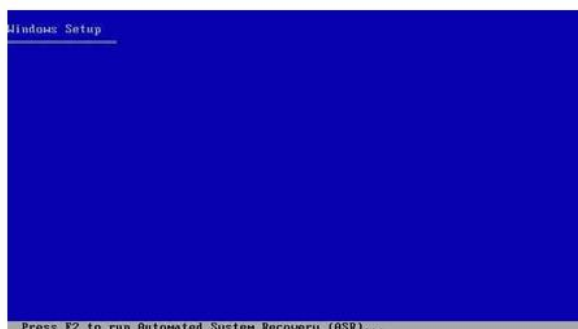


	Home Basic – "Vista Capable"	Home Premium / Business / Ultimate – "Vista Premium Ready"
Processor	800 MHz ^[50]	1.0 GHz
Memory	512 MB RAM	1 GB RAM
Graphics card	DirectX 9 capable	DirectX 9 capable GPU with Hardware Pixel Shader v2.0 and WDDM 1.0 driver support
Graphics memory	32MB RAM	128 MB RAM supports up to 2,756,000 total pixels (e.g. 1920 × 1200) or 512 MB+ for greater resolutions such as 2560x1600 ^[51]
HDD capacity	20 GB	40 GB
HDD free space	15 GB	15 GB
Other drives	CD-ROM	DVD-ROM
Audio	Audio output	Audio output

How To Install Windows XP

ASR

- Automated System Recovery
- The ASR process allows to restore the system disk (usually the C: drive) including the Windows files, all Registry settings and all user programs and data, allowing to recover a completely crashed system.
- During the restore process all data previously found on the System partition (usually C:) will be **erased**, then XP/2003 will be reinstalled, and the data from the ASR backup will be restored.



Create the ASR backup

1. Run NTBACKUP.EXE from the Run command or from the Program Files > Accessories > System Tools.
2. When the program starts click on the ASR button to invoke it's wizard.
3. The Wizard will start with a Welcome screen, advising that a Backup and the ASR disk will be created. Click Next.
4. Choose a destination and filename for the ASR backup. Make sure you do NOT select A: as the destination (duh...) nor C:, and that you have at least 2GB of free space on that partition.
5. The "Completing the ASR Preparation Wizard" will appear. When you click Finish, the Wizard creates a backup of your system files. You will then be asked to insert a floppy disk. You will later use this disk and the backup to restore your System in an event of a major failure.
6. To close this wizard and to begin the backup, click Finish".
7. The system will create a list of all files and then start the backup processes, creating a disk-backup-file. Once the disk-backup is done, you will be prompted for a floppy disk (blank, formatted) to store some ASR-data.

Use ASR to recover your system

1. Insert your Windows XP Professional Installation CD-ROM and boot from CD-ROM. If you get the message to "press any key to boot from CD...", press any key.
2. As soon as you see "Press F2 to run Automated System Recovery (ASR) ..." on the bottom line - press F2. You only have about 2 seconds to do so, so be prepared.
3. You will then be prompted to insert the ASR floppy disk. Do so.
You have for a few seconds still the time to abort the ASR process by pressing ESC.
4. Warning! The system will now start to format your C: drive. When it's done it will start with a process very similar to the XP installation.
5. After the file copying is done, the system will reboot and continue with the installation process, but it will not continue a complete installation. It starts the Automated System Recovery Wizard. This screen will wait for 90 sec, then it will continue automatically.
6. A screen will appear, asking you to confirm the location of the disk backup-file.
7. At the end of the ASR process, NTBACKUP.EXE is automatically used to restore the information on the C: drive.
8. After the process is over check to see if all data and settings have been restored properly.

Available commands in the Recovery Console

- **Attrib** Changes the attributes of a file or directory
- Batch** Executes the commands specified in the text file
- Bootcfg** Boot file (boot.ini) configuration and recovery
- ChDir** (Cd) Displays the name of the current directory or changes the current directory
- Chkdsk** Checks a disk and displays a status report
- Cls** Clears the screen
- Copy** Copies a single file to another location
- Delete** (Del) Deletes one or more files
- Dir** Displays a list of files and subdirectories in a directory
- Disable** Disables a system service or a device driver
- Diskpart** Manages partitions on your hard drives
- Enable** Starts or enables a system service or a device driver
- Exit** Exits the Recovery Console and restarts your computer
- Expand** Extracts a file from a compressed file
- Fixboot** Writes a new partition boot sector onto the specified partition
- Fixmbr** Repairs the master boot record of the specified disk
- Format** Formats a disk
- Help** Displays a list of the commands you can use in the Recovery Console
- Listsvc** Lists the services and drivers available on the computer
- Logon** Logs on to a Windows installation
- Map** Displays the drive letter mappings
- Mkdir** (Md) Creates a directory
- More** Displays a text file
- Rename** (Ren) Renames a single file
- Rmdir** (Rd) Deletes a directory
- Set** Displays and sets environment variables
- Systemroot** Sets the current directory to the systemroot directory of the system you are currently logged on to.
- Type** Displays a text file

